Advanced Topics in ACP

Lecture 2: The simply typed λ-calculus and the Curry-Howard correspondence

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Outline

1. Natural deduction for minimal logic, revisited
2. The Curry-Howard correspondence, informally
3. The simply typed λ-calculus
4. Conclusion, exercises and bibliography

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ND revisited

Def. We fix an infinite set of labels ranged over by $x, y, z, \ldots$. A labeled formula is a pair (x, A), also denoted by $x : A$ or A^x, where A is a formula and x is a label.
A set Γ of labeled formulas is coherent if $A = B$ whenever $A^x, B^x \in \Gamma$.

Def. A derivation $\mathcal{D}$ in ND with hypotheses $h(\mathcal{D})$ and conclusion $c(\mathcal{D})$ is:

- either A^x for any formula A and any label x, with $h(\mathcal{D}) = \{A^x\}$ and $c(\mathcal{D}) = A$;
- or it is obtained from derivations $\mathcal{D}'$, $\mathcal{D}_1$, $\mathcal{D}_2$ by applying one of the inference rules

$\Rightarrow$ elimination

$$\dfrac{\begin{array}{c}\vdots\,\mathcal{D}_1\\ A \Rightarrow B\end{array} \quad \begin{array}{c}\vdots\,\mathcal{D}_2\\ A\end{array}}{B}\ \Rightarrow_e$$

provided that $h(\mathcal{D}_1) \cup h(\mathcal{D}_2)$ is coherent

with $h(\mathcal{D}) = h(\mathcal{D}_1) \cup h(\mathcal{D}_2)$ and $c(\mathcal{D}) = B$

$\Rightarrow$ introduction

$$\dfrac{\begin{array}{c}[A^x]\\ \vdots\,\mathcal{D}'\\ B\end{array}}{A \Rightarrow B}\ \Rightarrow^x_i$$

provided that *all* occurrences of A^x in $\mathcal{D}'$ are discarded by $\Rightarrow^x_i$

with $h(\mathcal{D}) = h(\mathcal{D}') \setminus \{A^x\}$ and $c(\mathcal{D}) = A \Rightarrow B$

Idea. The label x determines how many occurrences of A are discharged by $\Rightarrow^x_i$.

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Examples of derivations in ND, revisited

Notation. $A_1^{x_1}, \ldots, A_n^{x_n} \vdash B$ (or $A_1^{x_1}, \ldots, A_n^{x_n} \vdash_{\mathsf{ND}} B$) means that there is a derivation in ND with conclusion B and hypotheses *among* the labeled formulas $A_1^{x_n}, \ldots, A_n^{x_n}$.

1. Prove that $A^x \vdash A$, and $\vdash A \Rightarrow A$, and $B^y \vdash A \Rightarrow A$, and $\vdash A \Rightarrow B \Rightarrow A$.
2. Give two distinct derivations of $A^x \vdash A \Rightarrow A$, and two distinct ones of $\vdash A \Rightarrow A \Rightarrow A$.
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An alternative presentation of ND via sequents: $\mathsf{ND}_{\mathsf{seq}}$, revisited

Def. An environment is a partial function with finite domain from labels to formulas. An environment Γ is denoted by $x_1 : A_1, \ldots, x_n : A_n$ if $\mathrm{dom}(\Gamma) = \{x_1, \ldots, x_n\}$ and $\Gamma(x_i) = A_i$.
A sequent is a pair $\Gamma \vdash A$ of an environment Γ and a formula A.

Def. A derivation in $\mathsf{ND}_{\mathsf{seq}}$ is a finite tree built up from the inference rules below.

$$\frac{}{\Gamma, x : A \vdash A}\,\mathsf{ax}^x \qquad \frac{\Gamma \vdash A \Rightarrow B \qquad \Gamma \vdash A}{\Gamma \vdash B} \Rightarrow_e \qquad \frac{\Gamma, x : A \vdash B}{\Gamma \vdash A \Rightarrow B} \Rightarrow_i^x$$

Notations. $\mathcal{D} \rhd_{\mathsf{ND}_{\mathsf{seq}}} \Gamma \vdash A$ means that $\mathcal{D}$ is a derivation in $\mathsf{ND}_{\mathsf{seq}}$ with conclusion $\Gamma \vdash A$.
$\Gamma \vdash_{\mathsf{ND}_{\mathsf{seq}}} A$ means that $\Gamma \vdash A$ is derivable in $\mathsf{ND}_{\mathsf{seq}}$ i.e. there is a derivation $\mathcal{D} \rhd_{\mathsf{ND}_{\mathsf{seq}}} \Gamma \vdash A$.

Idea. In $\mathsf{ND}_{\mathsf{seq}}$ a derivation makes explicit:

1. the current hypotheses at each inference step;
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Rmk. $x : X, y : X$ is an environment, while $x : X, x : X \Rightarrow X$ is not an environment!

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An alternative presentation of ND via sequents: $\mathsf{ND}_{\mathsf{seq}}$, revisited

Def. An environment is a partial function with finite domain from labels to formulas. An environment Γ is denoted by $x_1 : A_1, \ldots, x_n : A_n$ if $\mathrm{dom}(\Gamma) = \{x_1, \ldots, x_n\}$ and $\Gamma(x_i) = A_i$.
A sequent is a pair $\Gamma \vdash A$ of an environment Γ and a formula A.

Def. A derivation in $\mathsf{ND}_{\mathsf{seq}}$ is a finite tree built up from the inference rules below.

$$\frac{}{\Gamma, x : A \vdash A}\,\mathrm{ax}^x \qquad \frac{\Gamma \vdash A \Rightarrow B \qquad \Gamma \vdash A}{\Gamma \vdash B} \Rightarrow_e \qquad \frac{\Gamma, x : A \vdash B}{\Gamma \vdash A \Rightarrow B} \Rightarrow_i^x$$

Notations. $\mathcal{D} \triangleright_{\mathsf{ND}_{\mathsf{seq}}} \Gamma \vdash A$ means that $\mathcal{D}$ is a derivation in $\mathsf{ND}_{\mathsf{seq}}$ with conclusion $\Gamma \vdash A$.
$\Gamma \vdash_{\mathsf{ND}_{\mathsf{seq}}} A$ means that $\Gamma \vdash A$ is derivable in $\mathsf{ND}_{\mathsf{seq}}$ i.e. there is a derivation $\mathcal{D} \triangleright_{\mathsf{ND}_{\mathsf{seq}}} \Gamma \vdash A$.

Idea. In $\mathsf{ND}_{\mathsf{seq}}$ a derivation makes explicit:

1. the current hypotheses at each inference step;
2. if two occurrences of the same formula should be treated separately (different labels).

Rmk. $x : X, y : X$ is an environment, while $x : X, x : X \Rightarrow X$ is not an environment!

Examples of derivations in $\mathrm{ND}_{\mathrm{seq}}$, revisited

1. Prove $x : A \vdash A$, and $\vdash A \Rightarrow A$, and $y : B \vdash A \Rightarrow A$, and $\vdash A \Rightarrow B \Rightarrow A$.
2. Give two distinct derivations of $x : A \vdash A \Rightarrow A$, and two distinct ones of $\vdash A \Rightarrow A \Rightarrow A$.
3. Prove $\vdash (A \Rightarrow B) \Rightarrow (B \Rightarrow C) \Rightarrow A \Rightarrow C$ and $\vdash (A \Rightarrow (B \Rightarrow C)) \Rightarrow (A \Rightarrow B) \Rightarrow (A \Rightarrow C)$.

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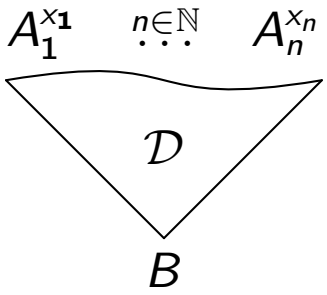
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Outline

1. Natural deduction for minimal logic, revisited
2. The Curry-Howard correspondence, informally
3. The simply typed λ-calculus
4. Conclusion, exercises and bibliography

A computational interpretation of ND

Idea. A derivation $\mathcal{D} \triangleright_{\mathsf{ND}} A_1^{x_1}, \dots, A_n^{x_n} \vdash B$ can be seen as a function t

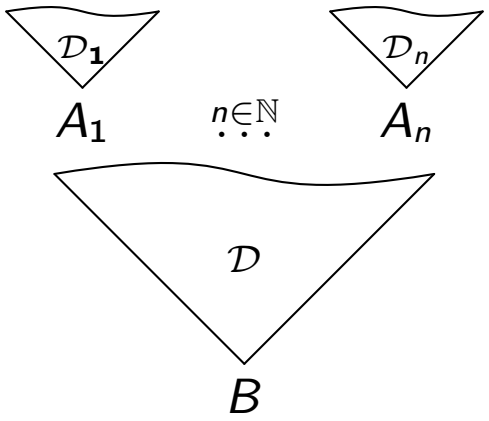


$$t \quad = \quad (x_1, \dots, x_n) \mapsto t(x_1, \dots, x_n)$$

that associates with derivations $\mathcal{D}_1 \triangleright_{\mathsf{ND}} \vdash A_1$, $\dots$, $\mathcal{D}_n \triangleright_{\mathsf{ND}} \vdash A_n$,



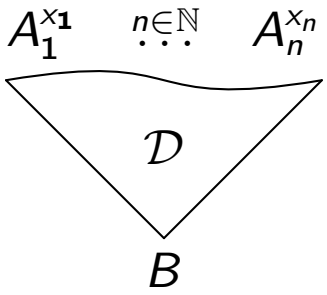
a derivation $t(\mathcal{D}_1, \dots, \mathcal{D}_n) \triangleright_{\mathsf{ND}} \vdash B$ obtained by composing $\mathcal{D}$ with $\mathcal{D}_1, \dots, \mathcal{D}_n$.



$\rightsquigarrow$ Let us see how, by induction on $\mathcal{D}$.

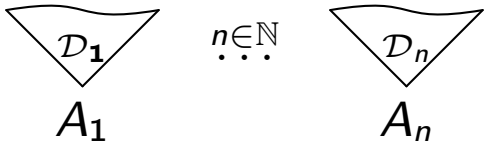
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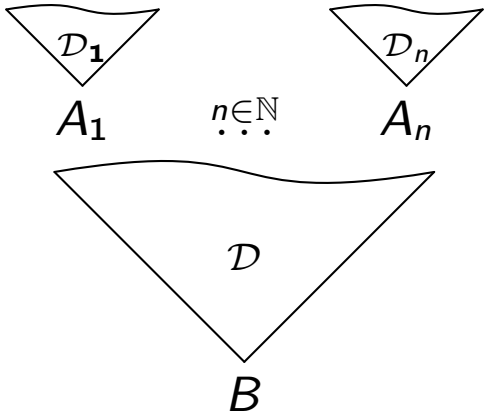


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A computational interpretation of ND: variable

Let the derivation $\mathcal{D} \triangleright_{\mathsf{ND}} A_1^{x_1}, \ldots, A_n^{x_n} \vdash B$ consist of a single hypothesis $A_i^{x_i}$, that is,

$$\mathcal{D} \quad = \quad A_i^{x_i} \qquad (\text{with } B = A_i)$$

The derivation $\mathcal{D}$ can then be seen a function

$$t \quad = \quad (x_1, \ldots, x_n) \mapsto x_i \qquad (\approx \text{projection})$$

represented by the variable x_i. Indeed, given the arguments $\mathcal{D}_1 \triangleright_{\mathsf{ND}} \vdash A_1, \ldots, \mathcal{D}_n \triangleright_{\mathsf{ND}} \vdash A_n$,

Rmk. In a derivation, different formulas are associated with different variables.
For several occurrences of A as hypotheses, we choose the same x or another variable.
$\rightsquigarrow$ A variable represents a (possibly empty) parcel of hypotheses of the same formula.

A computational interpretation of ND: variable

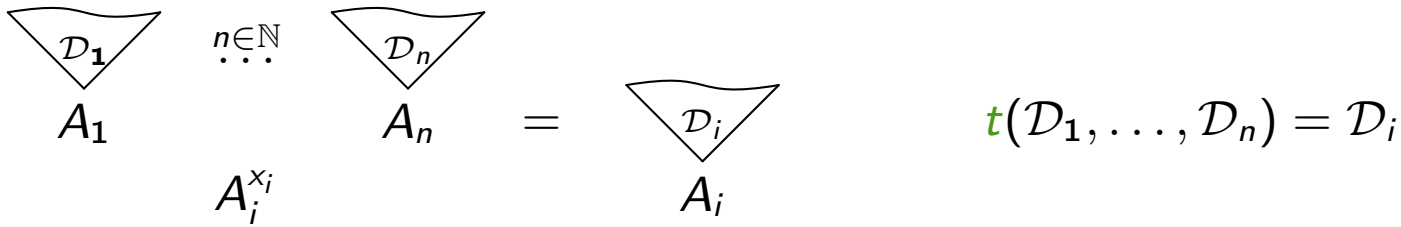
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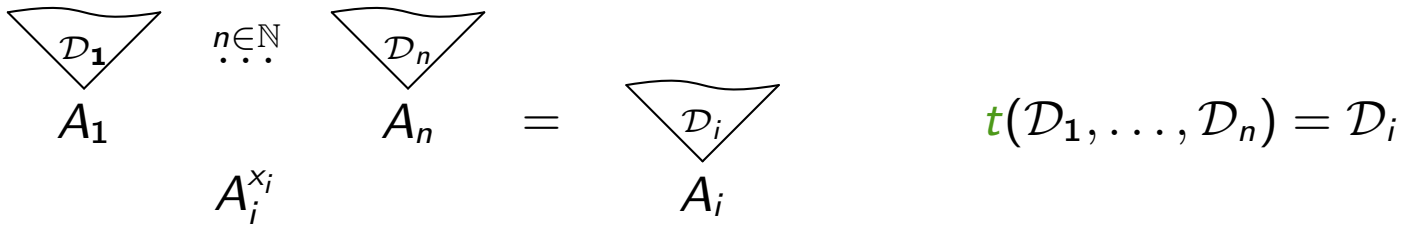
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A computational interpretation of ND: abstraction

Let the last rule of the derivation $\mathcal{D} \triangleright_{\mathsf{ND}} A_1^{x_1}, \ldots, A_n^{x_n} \vdash B$ be $\Rightarrow_i^y$, that is,

$$\mathcal{D} = \dfrac{\begin{array}{ccccc}[C^y] & A_1^{x_1} & n\in\mathbb{N} & A_n^{x_n} \\ & \multicolumn{3}{c}{\mathcal{D}'} \\ & \multicolumn{3}{c}{A}\end{array}}{C \Rightarrow A}\Rightarrow_i^y$$

(with $B = C \Rightarrow A$). Let

$$s \quad = \quad (y, x_1, \ldots, x_n) \mapsto s(y, x_1, \ldots, x_n)$$

be the function associated with the derivation $\mathcal{D}'$.

The derivation $\mathcal{D}$ can be seen as the function

$$t \quad = \quad (x_1, \ldots, x_n) \mapsto (\overbrace{y \mapsto s(y, x_1, \ldots, x_n)}^{(\lambda y.s)(x_1,\ldots,x_n)}) \qquad (\approx \text{currying})$$

represented by the abstraction $\lambda y.s$. Indeed, given $\mathcal{D}_1 \triangleright_{\mathsf{ND}} \vdash A_1, \ldots, \mathcal{D}_n \triangleright_{\mathsf{ND}} \vdash A_n$,

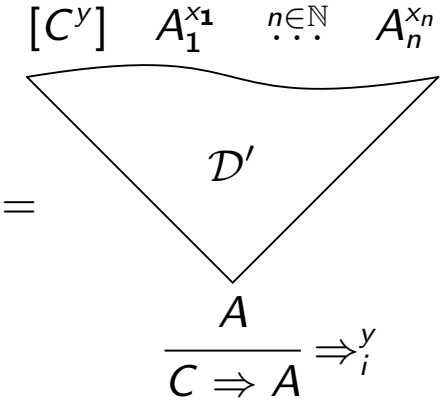
$$\dfrac{\begin{array}{ccccc}[C^y] & A_1^{x_1} & n\in\mathbb{N} & A_n^{x_n} \\ & \multicolumn{3}{c}{\mathcal{D}'} \\ & \multicolumn{3}{c}{A}\end{array}}{C \Rightarrow A}\Rightarrow_i^y \quad = \quad \dfrac{\begin{array}{ccccc}\mathcal{D}_1 & & n\in\mathbb{N} & & \mathcal{D}_n \\ A_1 & & & & A_n \\ [C^y] & & & & \\ & \multicolumn{3}{c}{\mathcal{D}'} \\ & \multicolumn{3}{c}{A}\end{array}}{C \Rightarrow A}\Rightarrow_i^y$$

$$t(\mathcal{D}_1, \ldots, \mathcal{D}_n) = (y \mapsto s(y, \mathcal{D}_1, \ldots, \mathcal{D}_n)) = \lambda y.s(y, \mathcal{D}_1, \ldots, \mathcal{D}_n)$$

(higher order)

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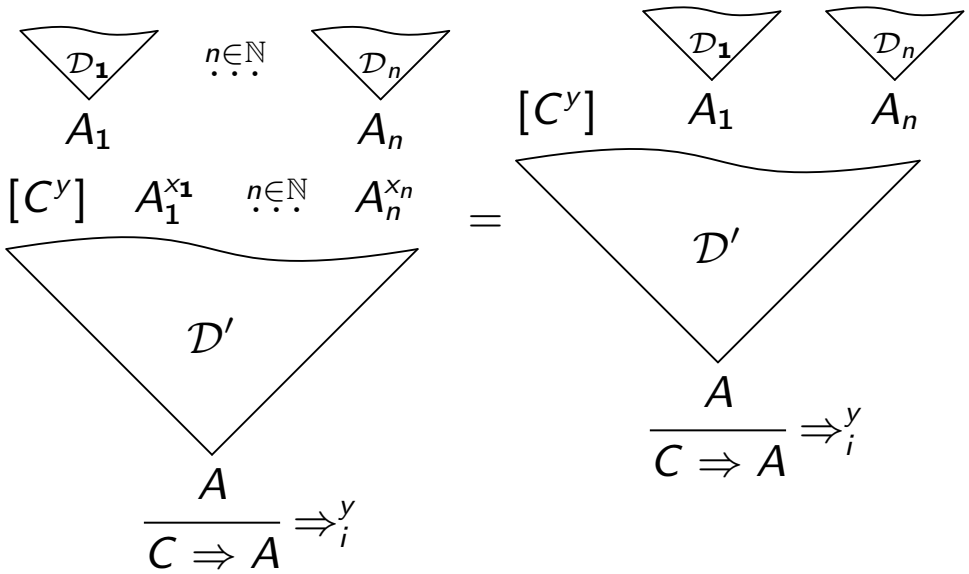
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A computational interpretation of ND: application

Let the last rule of the derivation $\mathcal{D} \triangleright_{\mathsf{ND}} A_1^{x_1}, \ldots, A_n^{x_n} \vdash B$ be $\Rightarrow_e$, that is,

$$\mathcal{D} = \dfrac{\dfrac{}{A \Rightarrow B}\,\mathcal{D}_1 \qquad \dfrac{}{A}\,\mathcal{D}_2}{B}\Rightarrow_e$$

Let s_1 and s_2 be the functions

$$s_1 = (x_1, \ldots, x_n) \mapsto s_1(x_1, \ldots, x_n)$$
$$s_2 = (x_1, \ldots, x_n) \mapsto s_2(x_1, \ldots, x_n)$$

associated with the derivations $\mathcal{D}_1$ and $\mathcal{D}_2$, respectively.

The derivation $\mathcal{D}$ can be seen as the function t applying s_1 to s_2

$$t = (x_1, \ldots, x_n) \mapsto s_1(x_1, \ldots, x_n)\big(s_2(x_1, \ldots, x_n)\big) \qquad (\approx \text{application})$$

represented by the juxtaposition $s_1 s_2$. Indeed, given $\mathcal{D}_1 \triangleright_{\mathsf{ND}} \vdash A_1, \ldots, \mathcal{D}_n \triangleright_{\mathsf{ND}} \vdash A_n$,

$$\dfrac{\dfrac{}{A \Rightarrow B}\,\mathcal{D}_1 \qquad \dfrac{}{A}\,\mathcal{D}_2}{B}\Rightarrow_e \;=\; \dfrac{\dfrac{}{A \Rightarrow B}\,\mathcal{D}_1 \qquad \dfrac{}{A}\,\mathcal{D}_2}{B}\Rightarrow_e$$

$$t(\mathcal{D}_1, \ldots, \mathcal{D}_n) = s_1(\mathcal{D}_1, \ldots, \mathcal{D}_n)\big(s_2(\mathcal{D}_1, \ldots, \mathcal{D}_n)\big)$$

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Let the last rule of the derivation $\mathcal{D} \triangleright_{\mathsf{ND}} A_1^{x_1}, \ldots, A_n^{x_n} \vdash B$ be $\Rightarrow_e$, that is,

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$$= \qquad \dfrac{\dfrac{}{A_1} \; \overset{n\in\mathbb{N}}{\cdots} \; \dfrac{}{A_n} \quad \mathcal{D}_1}{A \Rightarrow B} \qquad \dfrac{\dfrac{}{A_1} \; \overset{n\in\mathbb{N}}{\cdots} \; \dfrac{}{A_n} \quad \mathcal{D}_2}{A} \qquad \Rightarrow_e$$

$$t(\mathcal{D}_1, \ldots, \mathcal{D}_n) = s_1(\mathcal{D}_1, \ldots, \mathcal{D}_n)\big(s_2(\mathcal{D}_1, \ldots, \mathcal{D}_n)\big)$$

Summing up: decorating derivations in ND with terms

Given an infinite set of variables $x, y, z, \ldots$, terms are built up on the grammar:

$$s, t ::= x \mid \lambda x.t \mid st$$

Each derivation $\mathcal{D}$ in ND is associated with a unique term $(\mathcal{D})_\lambda$ defined inductively by:

$$(A^x)_\lambda = x \qquad \left(\dfrac{\begin{array}{c}[A^x]\\ \vdots\,\mathcal{D}\\ B\end{array}}{A \Rightarrow B}\Rightarrow_i^x\right)_\lambda = \lambda x.(\mathcal{D})_\lambda \qquad \left(\dfrac{\begin{array}{cc}\vdots\,\mathcal{D}_1 & \vdots\,\mathcal{D}_2\\ A \Rightarrow B & A\end{array}}{B}\Rightarrow_e\right)_\lambda = (\mathcal{D}_1)_\lambda(\mathcal{D}_2)_\lambda$$

Rmk. The variable labeling the hypotheses and $\Rightarrow_i$ is crucial to uniquely determine $(\mathcal{D})_\lambda$.

We have a3 static correspondence between derivations in ND and terms.

- We can decorate each formula occurrence in a derivation in ND with a term.
- For every derivation $\mathcal{D}$ in ND, its term $(\mathcal{D})_\lambda$ is the decoration of its conclusion.
- Terms are a concise (and linear) representations of derivations (trees) in ND.

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Dynamics: cut-elimination as a computational step

$\mathcal{D}_0 =$

$$\dfrac{\dfrac{\begin{array}{c}y : [A^y] \\ \vdots \\ t(y,\vec{x}) : B\end{array}}{\lambda y.t(y,\vec{x}) : A \Rightarrow B}\Rightarrow_i^y \quad \begin{array}{c}\mathcal{D}' \\ s(\vec{x}) : A\end{array}}{(\lambda y.t(y,\vec{x}))s(\vec{x}) : B}\Rightarrow_e$$

$\vdots$

$\rightarrow_{\mathsf{cut}}$

$$\begin{array}{c}\mathcal{D}' \quad\quad \mathcal{D}' \\ s(\vec{x}) : A \quad \overset{n\in\mathbb{N}}{\vdots} \quad s(\vec{x}) : A \\ \hline \mathcal{D} \\ \hline t(s(\vec{x}),\vec{x}) : B \\ \vdots\end{array} = \mathcal{D}_1$$

Rmk. Cut-elimination explains why term juxtaposition can be seen as function application.

The decoration of derivations with terms commutes with cut-elimination via the step:

$$(\lambda y.t)s \rightarrow_\beta t\{s/y\}$$

where $t\{s/y\}$ stands for the substitution of s for the free occurrences of y in t.

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Examples of decorations of derivations in ND

$$x : A$$

$$\frac{x : [A^x]}{\lambda x.x : A \Rightarrow A}\Rightarrow_i^x$$

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Example of decorations of derivations in ND with cut-elimination

$$\dfrac{\dfrac{\dfrac{x : [(X \Rightarrow X)^x]}{\lambda x.x : (X \Rightarrow X) \Rightarrow X \Rightarrow X}\Rightarrow_i^x \quad \dfrac{y : [X^y]}{\lambda y.y : X \Rightarrow X}\Rightarrow_i^y}{(\lambda x.x)\lambda y.y : X \Rightarrow X}\Rightarrow_e}{\rightarrow_{\text{cut}} \quad \dfrac{y : [X^y]}{\lambda y.y : X \Rightarrow X}\Rightarrow_i^y}$$

Rmk. $(\lambda x.x)\lambda y.y \rightarrow_\beta x\{\lambda y.y/x\} = \lambda y.y \rightsquigarrow$ cut-elimination commutes with decoration.

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$\downarrow$cut

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Rmk. $(\lambda x.\lambda y.\lambda z.(xz)(yz))\lambda a.\lambda b.a \to_\beta (\lambda y.\lambda z.(xz)(yz))\{\lambda a.\lambda b.a/x\} =$
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Curry-Howard and inverse decoration: from terms to derivations

So far, we have seen a first taste of the Curry-Howard correspondence:

1. derivations in ND (and $\mathsf{ND}_{\mathsf{seq}}$) can be decorated with λ-terms;
2. β-reduction (transforming λ-terms) mimics cut-elimination steps and vice versa.

Can we invert the process? Can derivations be associated with λ-terms?

Question. Given the term $\lambda f.\lambda x.fx$, what is the derivation associated with it?

$$\dfrac{\dfrac{\dfrac{f : [?^f] \qquad x : [?^x]}{fx :?}\Rightarrow_e}{\lambda x.fx :?}\Rightarrow_i^x}{\lambda f.\lambda x.fx :?}\Rightarrow_i^f$$

Problem. Without knowing the formulas associated with variables, there is no answer.

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Rmk. Fixing formulas for variables (and hence for the whole term) is crucial!

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Outline

1. Natural deduction for minimal logic, revisited
2. The Curry-Howard correspondence, informally
3. The simply typed λ-calculus
4. Conclusion, exercises and bibliography

The simply typed λ-calculus in Curry-style: typing rules

Types: $A, B ::= X \mid A \Rightarrow B$ (given a set of ground types ranged over by $X, Y, Z \dots$)

(λ-)Terms: $s, t ::= x \mid \lambda x.t \mid st$ (called variable, abstraction, application, respectively)

Def. An environment is a partial function with finite domain from variables to types. An environment Γ is denoted by $x_1 : A_1, \dots, x_n : A_n$ if $\mathrm{dom}(\Gamma) = \{x_1, \dots, x_n\}$ and $\Gamma(x_i) = A_i$.

A judgment is a triple $\Gamma \vdash t : A$ where Γ is an environment, t is a term, A is a type.

Def. A derivation in STLC is a finite tree built up from the typing rules below.

$$\frac{}{\Gamma, x : A \vdash x : A}\,\mathrm{var} \qquad \frac{\Gamma, x : A \vdash t : B}{\Gamma \vdash \lambda x.t : A \Rightarrow B}\,\lambda \qquad \frac{\Gamma \vdash s : B \Rightarrow A \quad \Gamma \vdash t : B}{\Gamma \vdash st : A}\,@$$

Notation: $\mathcal{D} \rhd_{\mathsf{STLC}} \Gamma \vdash t : A$ means that $\mathcal{D}$ is a derivation in STLC with conclusion $\Gamma \vdash t : A$. $\Gamma \vdash_{\mathsf{STLC}} t : A$ means that there is a derivation $\mathcal{D} \rhd_{\mathsf{STLC}} \Gamma \vdash t : A$ in STLC; we then say that $\Gamma \vdash t : A$ is derivable in STLC. A term t is typable in STLC if $\Gamma \vdash_{\mathsf{STLC}} t : A$ for some Γ, A.

Ex. The terms x, $\lambda x.x$, xy and $\lambda x.\lambda y.xy$ are typable in STLC; xx and $\lambda x.xx$ are not.

Rmk. Types are the formulas of minimal logic. Derivations in STLC are the ones in $\mathsf{ND}_{\mathsf{seq}}$ plus a decorating λ-term. A redex in $\mathsf{ND}_{\mathsf{seq}}$ corresponds to a β-redex in their λ-term.

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The simply typed λ-calculus in Curry-style: free variables and β-reduction

In $\lambda x.t$, the λx is a binder for the occurrences of x in t not in the scope of another λx.

Def. The free variables of a term t are the variables that are not bound to a λ. Formally,

$$\mathrm{fv}(x) = \{x\} \qquad \mathrm{fv}(st) = \mathrm{fv}(s) \cup \mathrm{fv}(t) \qquad \mathrm{fv}(\lambda x.t) = \mathrm{fv}(t) \setminus \{x\}.$$

Rmk. Let $t = (z(\lambda x.\lambda y.xy))x$. Then $\mathrm{fv}(t) = \{x, z\}$. Note that the red occurrence of x is bound to λx, but the blue occurrence of x is unbound, thus $x \in \mathrm{fv}(t)$.

Proposition

1. If $\Gamma \vdash_{\mathsf{STLC}} t : A$, then $\Gamma, x : B \vdash_{\mathsf{STLC}} t : A$, for any type B and variable $x \notin \mathrm{dom}(\Gamma)$.
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Moral. If $\Gamma \vdash_{\mathsf{STLC}} t : A$, then Γ is essentially a type assignment for $\mathrm{fv}(t)$.

β-reduction $\to_\beta$ is a binary relation on λ-terms generated by:

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2. If $\Gamma \vdash_{\mathsf{STLC}} t : A$, then $\mathrm{fv}(t) \subseteq \mathrm{dom}(\Gamma)$ and $\Gamma\restriction_{\mathrm{fv}(t)} \vdash_{\mathsf{STLC}} t : A$.

Moral. If $\Gamma \vdash_{\mathsf{STLC}} t : A$, then Γ is essentially a type assignment for $\mathrm{fv}(t)$.

β-reduction $\to_\beta$ is a binary relation on λ-terms generated by:

$$(\lambda x.t)s \to_\beta t\{s/x\}$$

where $t\{s/x\}$ is the capture-avoiding substitution of s for the free occurrences of x in t.

The simply typed λ-calculus in Curry-style: free variables and β-reduction

In $\lambda x.t$, the λx is a binder for the occurrences of x in t not in the scope of another λx.

Def. The free variables of a term t are the variables that are not bound to a λ. Formally,

$$\mathrm{fv}(x) = \{x\} \qquad \mathrm{fv}(st) = \mathrm{fv}(s) \cup \mathrm{fv}(t) \qquad \mathrm{fv}(\lambda x.t) = \mathrm{fv}(t) \setminus \{x\}.$$

Rmk. Let $t = (z(\lambda x.\lambda y.xy))x$. Then $\mathrm{fv}(t) = \{x, z\}$. Note that the red occurrence of x is bound to λx, but the blue occurrence of x is unbound, thus $x \in \mathrm{fv}(t)$.

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The capture-avoiding substitution

Naive substitution $t[s/x]$: replacement of the free occurrences of the variable x in t by s.

Ex: Let $t = \lambda y.yx$ and $s = yy$. Then, $t[s/x] = \lambda y.y(yy)$.
Problem: The free variable y in s has been captured by the λ in t. $\rightsquigarrow$ Undesirable.

Solution: Capture-avoiding substitution $t\{s/x\}$

1. first rename the bound variables in t with variables that do not occur in t or s;
2. then perform the substitution in of s for x in t.

$\rightsquigarrow$ So, the free variables of s are not captured by the λ's in t.

Ex: Let $t = \lambda y.yx$ and $s = yy$. Then, $t\{s/x\} = \lambda z.z(yy) = \lambda x.x(yy) \neq \lambda y.y(yy)$.

Rmk: The operation of renaming the bound variables in a term is called α-equivalence.
$\rightsquigarrow$ Capture-avoiding substitution makes sense, as we identify terms up to α-equivalence.

Idea. A bound variable occurrence is muted, it lose its identity and can be renamed.

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Some remarks about the simply typed λ-calculus in Curry-style

Rmk. The search for a derivation is uniquely determined by the term (syntax-directed).
$\rightsquigarrow$ To build a derivation $\mathcal{D}$ of $\Gamma \vdash t : A$, just look at t to know the last rule of $\mathcal{D}$ (if any).

The types used in the simply typed λ-calculus are exactly the formulas of minimal logic.
The inference rules for the simply typed λ-calculus are the ones of $\mathrm{ND}_{\mathrm{seq}}$ plus decoration.
$\rightsquigarrow$ Every derivation in $\mathrm{ND}/\mathrm{ND}_{\mathrm{seq}}$ corresponds to a unique λ-term typed in Curry-style.

Question: With every typable term in Curry-style is it associated a unique derivation? No!

$$\dfrac{\dfrac{}{x : X \vdash x : X}\,\mathrm{var}}{\vdash \lambda x.x : X \Rightarrow X}\,\lambda \qquad\qquad \dfrac{\dfrac{}{x : X \Rightarrow X \vdash x : X \Rightarrow X}\,\mathrm{var}}{\vdash \lambda x.x : (X \Rightarrow X) \Rightarrow X \Rightarrow X}\,\lambda$$

$\rightsquigarrow$ The map from $\mathrm{ND}/\mathrm{ND}_{\mathrm{seq}}$ derivations to typable terms in Curry-style is not injective!

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Idea. Let us make types intrinsic to terms (static typing, *a priori*): Church-style.

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Derivability of a judgment and typability of a λ-term are defined as in Curry-style.

β-reduction ($t\{s/x\}$ is the capture-avoiding substitution of s for the free occurrences of x in t):

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Rmk. Syntax-directedness and proposition on p. 21 hold true in Church-style as well.

Notations. $\mathcal{D} \rhd_{\text{Curry/Church}} \Gamma \vdash t : A$ means that $\mathcal{D} \rhd_{\text{STLC}} \Gamma \vdash t : A$ in Curry/Church-style.
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Curry-style versus Church-style

Church-style terms are related to Curry-style terms by the forgetful function $\lceil \cdot \rceil$:

$$\lceil x \rceil = x \qquad \lceil \lambda x^A.t \rceil = \lambda x.\lceil t \rceil \qquad \lceil st \rceil = \lceil s \rceil \lceil t \rceil$$

Proposition

1. If $\Gamma \vdash_{\text{Church}} t : A$, then $\Gamma \vdash_{\text{Curry}} \lceil t \rceil : A$.
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Proof. By induction on the derivation in Church (Point 1) or Curry (Point 2) style. □

Rmk: $\lambda x^X.x$ and $\lambda x^{X \Rightarrow X}.x$ are different terms in Church-style, because $X \neq X \Rightarrow X$.

Proposition (Uniqueness of type and derivation for typable terms in Church-style)

If $\mathcal{D} \rhd_{\text{Church}} \Gamma \vdash t : A$ and $\mathcal{D}' \rhd_{\text{Church}} \Gamma \vdash t : A'$, then $A = A'$ and $\mathcal{D} = \mathcal{D}'$.

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Some properties of the simply typed λ-calculus (Curry and Church style)

Lemma (Substitution)

If $\Gamma, x : B \vdash t : A$ and $\Gamma \vdash s : B$ are derivable, then so is $\Gamma \vdash t\{s/x\} : A$.

Proof. By structural induction on t (exercise!). □

Theorem (Subject reduction)

If $\Gamma \vdash_{\mathsf{STLC}} t : A$ and $t \to_\beta s$, then $\Gamma \vdash_{\mathsf{STLC}} s : A$.

Proof. By structural induction on t, using the substitution lemma in the key-case. □

Rmk. The converse (subject expansion) does not hold: let $A = Y \Rightarrow X \Rightarrow X$, $s = \lambda y.\lambda z.z$ and $t = (\lambda x.\lambda y.\lambda z.(xz)(yz))\lambda x.\lambda y.x$, then $t \to^*_\beta s$ and $\vdash_{\mathsf{STLC}} s : A$, but $\not\vdash_{\mathsf{STLC}} t : A$.

Theorem (Normalization)

If $\Gamma \vdash_{\mathsf{STLC}} t : A$, then there is $\mathcal{D} \rhd_{\mathsf{STLC}} \Gamma \vdash s : A$ without redexes such that $t \to^*_\beta s$.

Proof. Exactly the proof of cut-elimination in ND (uppermost in ND $\rightsquigarrow$ innermost in λ). □

Some properties of the simply typed λ-calculus (Curry and Church style)

Lemma (Substitution)

If $\Gamma, x : B \vdash t : A$ and $\Gamma \vdash s : B$ are derivable, then so is $\Gamma \vdash t\{s/x\} : A$.

Proof. By structural induction on t (exercise!). □

Theorem (Subject reduction)

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The Curry-Howard correspondence in a nutshell

minimal logic	simply typed λ-calculus	computer science
formula	type	specification
derivation	term	program
cut-elimination step	β-reduction	computation step
derivation without redexes	normal form	result
cut-elimination theorem	normalization	termination

Concerning the correspondence between derivations and terms:

derivation in minimal logic	term in simply typed λ-calculus
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$\Rightarrow_i$	abstraction λ
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Outline

1. Natural deduction for minimal logic, revisited
2. The Curry-Howard correspondence, informally
3. The simply typed λ-calculus
4. Conclusion, exercises and bibliography

What have we learned today?

1. How to decorate derivations in natural deduction for minimal logic with λ-terms.
2. The procedure of β-reduction on λ-terms.
3. Church and Curry styles for the simply typed λ-calculus (STLC).
4. The Curry–Howard correspondence between ND for minimal logic and STLC.
5. Some properties of the simply typed λ-calculus (subject reduction, normalization).

Questions?

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Non-assessed exercises

1. Find the simply typed λ-terms (in Curry-style and Church-style) associated with the derivations in ND found for the facts below (see Exercise 1 from Lecture 1).
   1. $\vdash X \Rightarrow ((X \Rightarrow Y) \Rightarrow Y)$.
   2. $((X \Rightarrow Y) \Rightarrow X \Rightarrow Z)^z \vdash Y \Rightarrow X \Rightarrow Z$.
   3. $((X \Rightarrow Y) \Rightarrow X)^z \vdash Y \Rightarrow X$.
   4. $(X \Rightarrow Y \Rightarrow Z)^z \vdash Y \Rightarrow X \Rightarrow Z$.
   5. $(X \Rightarrow Y \Rightarrow Z)^z, (X \Rightarrow Y)^y \vdash X \Rightarrow Z$.
   6. $((X \Rightarrow X) \Rightarrow Y)^y \vdash (Y \Rightarrow Z) \Rightarrow Z$.
2. Perform all possible β-reduction steps from the λ-term decorating the derivation $\mathcal{D}$ in ND on p. 31 of Lecture 1, until you get a β-normal form. Is it always the same? Compare it with the normal derivation obtained by cut-elimination steps from $\mathcal{D}$.
3. Prove rigorously the following facts ($f^n x = \overbrace{f(\dots(f\,x)\dots)}^{n \text{ times } f}$ for any $n \in \mathbb{N}$):
   1. $\lambda x.xx$ is untypable in Curry-style, $\lambda x^A.xx$ is untypable in Church-style for any type A;
   2. in Church-style, $\lambda f^Y.\lambda x^X.f^n x$ is not typable for any $n > 0$ but $\lambda f^Y.\lambda x^X.x$ is typable;
   3. $\lambda f.\lambda x.f^n x$ is typable in Curry-style, for all $n \in \mathbb{N}$.
4. Prove that if t is typable in Church or Curry style, then so is every subterm of t.
5. Prove rigorously the propositions on p. 21 and 25, the lemma and theorems on p. 26.
6. Let $A\{B/X\}$ be the type obtained from the type A by substituting B for each occurrence of the ground type X. Let $\Gamma\{A/X\}$ be its generalization to environments. Show that if $\Gamma \vdash t : A$ is derivable in Curry-style, then so is $\Gamma\{B/X\} \vdash t : A\{B/X\}$.
7. Is the previous point valid in Church-style? What change is needed to make it true?
8. Prove that $(\lambda x.(\lambda y.y))\lambda z.zz$ is not typable (in Curry-style). Deduce that subject expansion (see p. 26) does not hold in the simply typed λ-calculus.

Bibliography

- For more about the simply typed λ-calculus:


Ralph Loader. *Notes on Simply Typed λ-Calculus*. Technical report ECS-LFCS-98-381, University of Edinburgh, 1998. `http://www.lfcs.inf.ed.ac.uk/reports/98/ECS-LFCS-98-381/ECS-LFCS-98-381.pdf`. [Chapters 1–3]


Henk Barendregt. *Lambda Calculi with Types*. In S. Abramsky et al. (eds), Handbook of Logic in Computer Science; vol. 2, 117-309, 1992. `https://repository.ubn.ru.nl/bitstream/handle/2066/17231/17231.pdf` [Chapter 3]

- For more about the Curry-Howard correspondence:


Jean-Yves Girard, Yves Lafont, Paul Taylor. *Proofs and Types*. Cambridge Tracts in Theoretical Computer Science, Vol. 7, Cambridge University Press, 1989. `https://www.paultaylor.eu/stable/prot.pdf`. [Chapters 2–4, 6]